Introduction to Quarto

Write, analyze, and publish in a single document

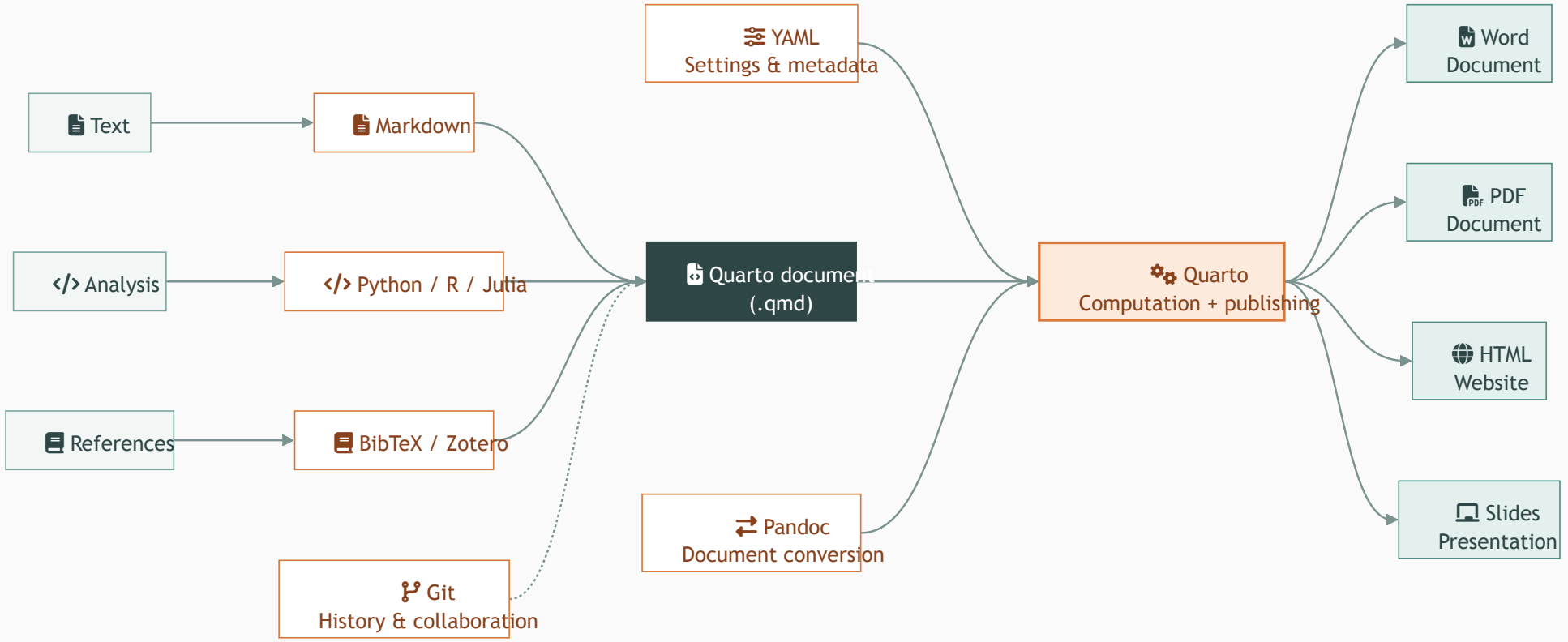
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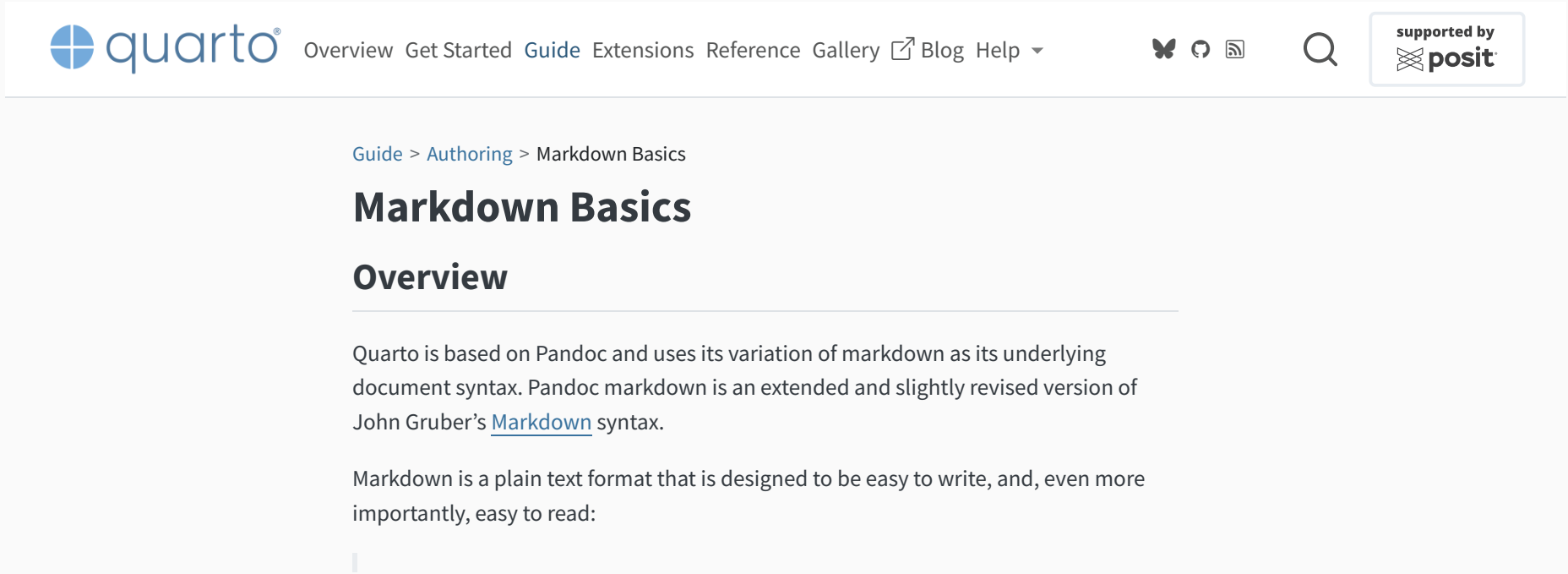
Why Quarto?

Why Quarto


YAML
Settings & metadata
Text
Markdown
Word Document
PDF Document
Analysis
Python / R / Julia
Quarto document (.qmd)
Quarto
Computation + publishing
HTML Website
References
BibTeX / Zotero
Slides Presentation
Pandoc
Document conversion
Git
History & collaboration

Markdown

- What? Lightweight markup for text formatting.
- Why? Keep the source readable and focus on content and structure.



Open Quarto Markdown Basics ↗

Pandoc

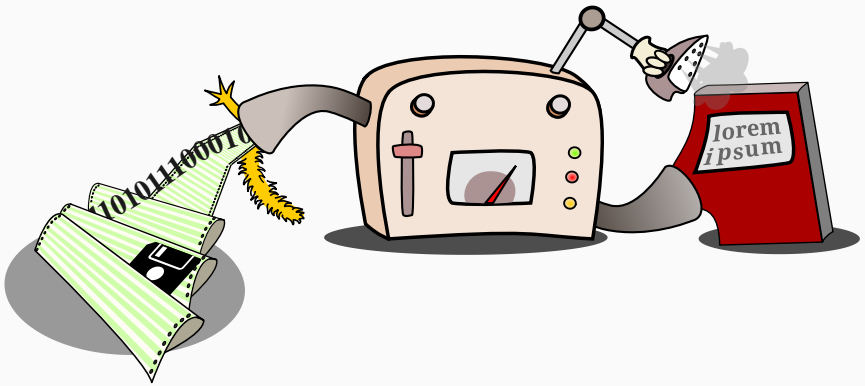
- What? Converts documents between formats such as Markdown, HTML and Word.
- Why? One source, many outputs: the conversion machinery inside Quarto.

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Pandoc a universal document converter

About Installing Demos ▼ Documentation Help Extras Releases

Search



If you need to convert files from one markup format into another, pandoc is your swiss-army knife. Pandoc can convert between the following formats:

(← = conversion from; → = conversion to; ↔ = conversion from and to)

Lightweight markup formats

Open the Pandoc website ↗

YAML

- What? A readable data format built from keys, values and indentation.
- Why? In Quarto: declare the document's metadata and publishing settings.

About YAML

What is YAML?

YAML (YAML Ain't Markup Language) is a human-readable data serialization language designed for data exchange between languages with different data structures.

It's often used for configuration files, data exchange, and representing complex data structures in a readable format.

Key Features

- **Human Readable** - Easy to read and write for both humans and machines
- **Language Independent** - Works across programming languages and platforr
- **Hierarchical** - Supports complex nested data structures
- **Extensible** - Can be extended with custom data types
- **Secure** - Built with security considerations in mind

Open the YAML website ↗

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Git

- What? A distributed version control system that records changes to files.
- Why? Keep a history, compare versions, and collaborate on shared work.

git --print-out

Git is a free and open source distributed version control system designed to handle everything from small to very large projects with speed and efficiency.

Git is lightning fast and has a huge ecosystem of GUIs, hosting services, and command-line tools.



About
Git's performance and ecosystem

Learn
Pro Git book, videos, tutorials, and cheat sheet

Latest source release
2.56.0
Release Notes (2026-09-28)

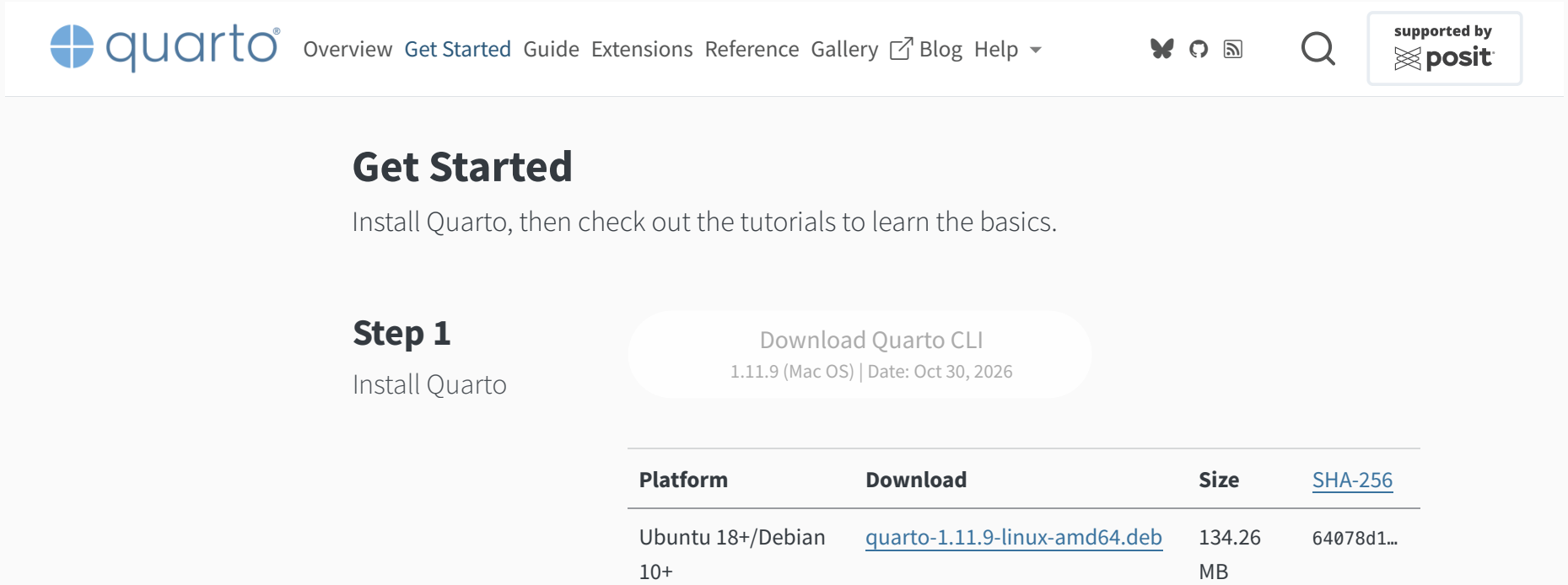
Open the Git website ↗

What is Quarto

- **Open-source scientific and technical publishing system**
  → Leverages modern tools for content and structure (Markdown) and configuration (YAML)
- **Favours reproducibility and transparency**
  → Supports executable code within text
- **Supports multiple output formats**
  → Pandoc is the conversion machinery underneath!
- **Successor to R Markdown but language-agnostic**
  → *Python*, *R*, *Julia*, ...
- **Integration in most IDE**
  → *VS Code*, *RStudio*, *Jupyter*...

Getting started

- Install Quarto + IDE extension
- Follow the tutorials



Open this page ↗

Objectives

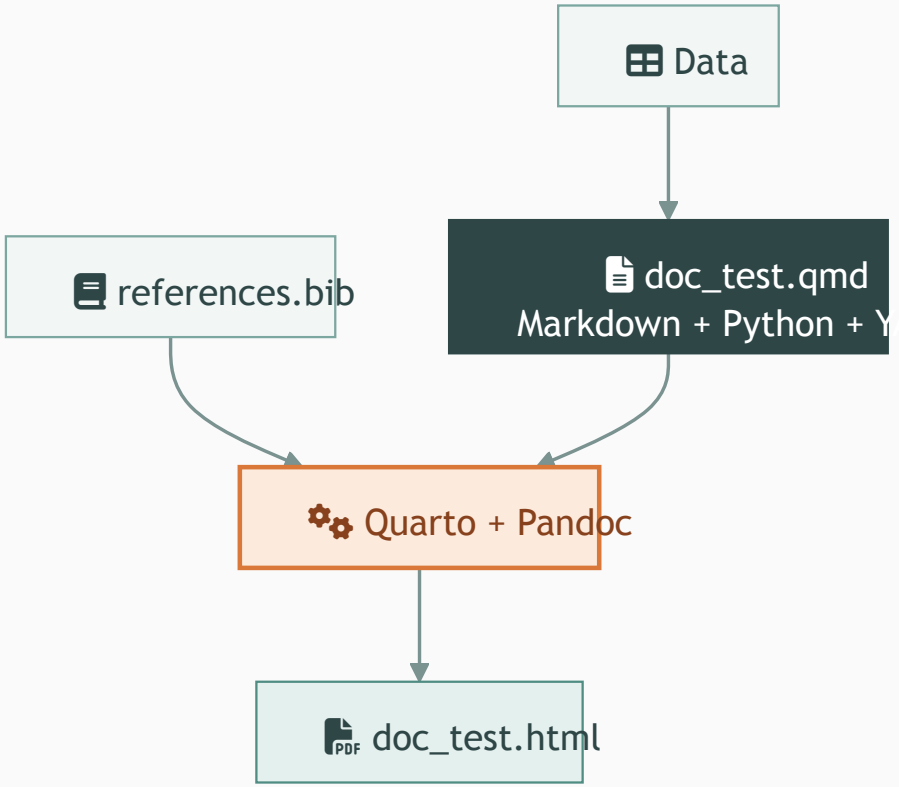
Aim: **Demonstrate the "Write, Analyze, Publish" workflow**

- **Part I: Quarto for research**
    1. Write a document from scratch
    2. Complex multi-output workflow
- **Part II: Quarto for teaching** - *my workflow* to:
    1. Write a textbook website
    2. Add interactive exercises
    3. Develop interactive slides

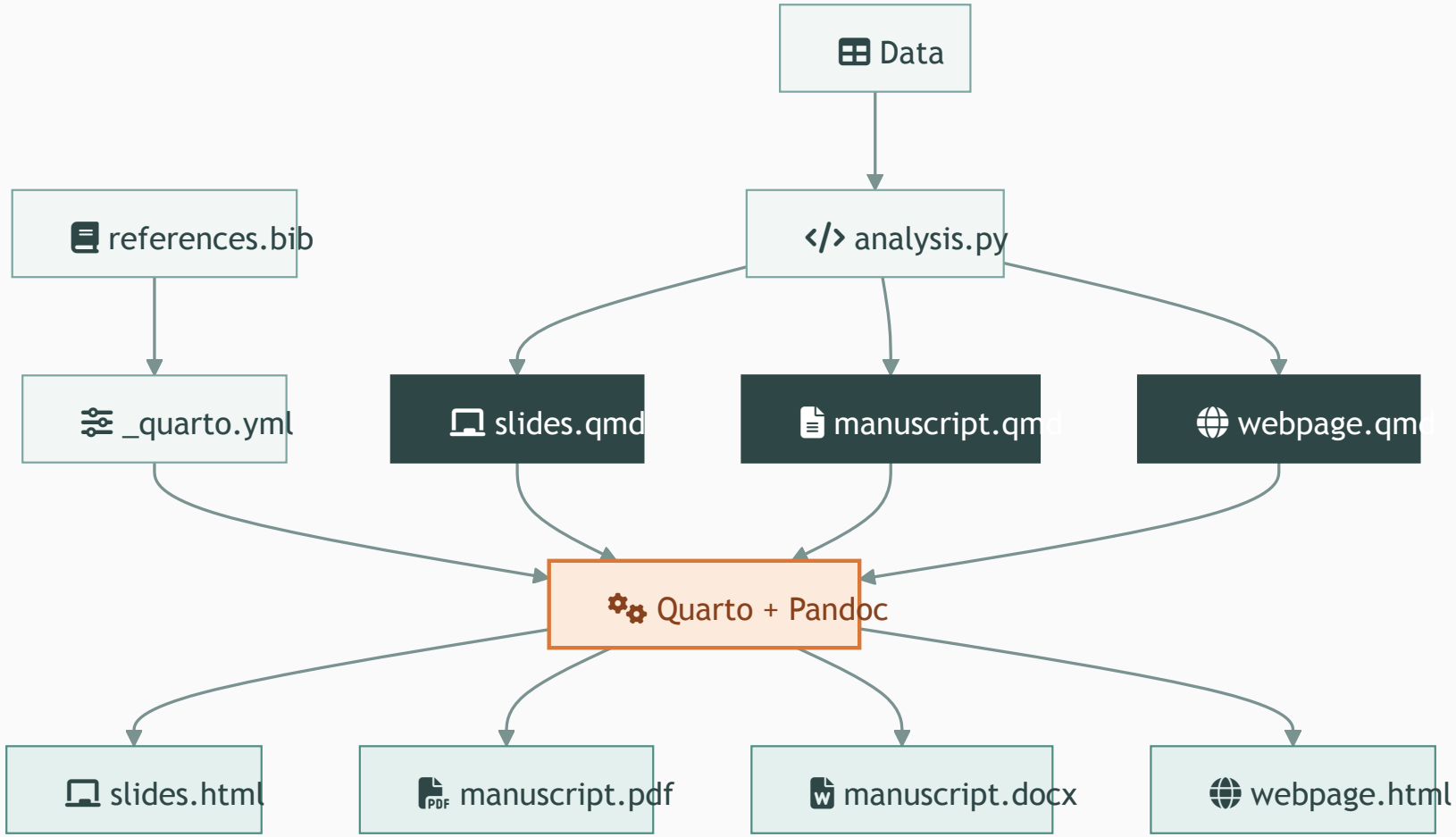
Quarto for research

One document, one output

- **One `.qmd` document**
  - → Text → Markdown
  - → Code → Python
  - → Configuration → YAML
- **Linked to**
  - → Dataset → remote
  - → Bibliography → references.bib
- **One output** → HTML



One analysis, several outputs



Quarto for teaching

Text book

- **Teaching book**
  - → Cross-referencing
  - → Collaborative tools
- **Exercises**
  - → Profiles for *students*, *lecturer* or *corrections*
  - → Interactive coding exercises
- **Slides**
  - → Drawing
  - → Code integration + animation
  - → Live coding
  - → Integration with medias
  - → Other cool tools

Now what?

Conclusion

... easy to get started

... learning curve to get into some more complex workflows

This presentation / examples



My teaching template



Examples

Volcanology

Outils informatiques

Data science